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Coastal Process Impact Assessment in relation to Offshore Wave Energy Development

Jimmy Murphy

Abstract

At present there is no commercial operational wave energy site worldwide but a number of countries with high resource levels have set up test sites with the view to deploy pre-commercial prototype devices. The creation of these sites has lead to the examination on the potential impacts on the coastal physical environment particularly since Wave Energy Convertors (WECs) can have a direct impact on the local and far field hydrodynamics. Potential changes to the wave and current regime could affect human activities (boating, surfing etc.) as well as sedimentary processes possibly leading to areas of erosion and accretion along the affected coastline. The magnitude of any impact depend on a number of factors which include; size and type of WEC, array layout, location relative to shoreline, wave climate , seabed conditions and coastline type. As it is likely that no two sites would be the same, it makes the development of a standard assessment methodology very difficult. While there are a number of environment concerns related to the development of an offshore renewable energy project this paper focuses on assessing the impacts on the physical environment. Other areas of concern such as ecological, biological, conservation, etc are not considered in this assessment.

Much of the work carried out thus far on offshore renewables impact assessment relate to wind turbine developments and whilst these provide useful information the methodologies developed cannot be completely applied to WEC sites. Generally for offshore wind developments apart from local effects around the foundation structure (primarily scour), the overall impact is negligible. Therefore in the current study existing site assessment methodologies are examined through a review of both offshore wind energy developments and the limited ocean energy deployments that have been undertaken thus far. Based on this review a revised methodology is developed for wave energy sites which is then applied to the Belmullet wave energy test site in Co. Mayo in Ireland.

Review of assessment procedure for Offshore Wind Farms

Although offshore wind farms do not have as strong an interaction with physical processes they still do have various impacts and for any development an assessment of these needs to be carried out which usually forms part of an EIA. Whilst the categorisation of impacts vary from project to project four general groupings are common, these relate to

- Construction - all possible impacts and environmental sensitivities in relation to the windfarm construction. The construction phase is usually includes everything prior to the wind farm becoming operational. Impacts in this phase usually consist of increased suspended sediment concentrations as a result of seabed preparation, foundation installation, cable laying and initial scouring around the foundation structure
- Operational - any impact whilst the wind farm is operational which is considered the case until decommissioning occurs. Potential impacts include - changes to currents and wave patterns due to the presence of foundation or any ancillary constructions, changes to the sediment transport pathways and changes to the erosion deposition patterns on the coastline. Although it is not known of any offshore wind farm where an issue in relation to erosion of the adjacent shoreline has arisen.
- Decommissioning - possible impacts caused during the decommissioning process. This stage lasts from the start of decommissioning until the whole farm is dismantled and removed from the site. Examples of potential impacts include - increased suspended sediment concentrations

- Cumulative refers to impacts of the development in tandem with other offshore developments such as other wind farms, oil platforms etc.

For wind farms potential impacts are usually categorised according to their severity. The following are the four main categories

- Major - Impacts of such a significance that an alteration of the proposed project is considered
- Moderate - Impacts of sufficient importance to consider introducing mitigation measures
- Minor - Impacts which are unlikely to be considered important to require mitigation measures
- Negligible - Impacts which are deemed to be of low importance and not considered in the decision making process

For example the scoping report for the wind farm extension at the Burbo Bank produced by Dong Energy classified the significance of all physical impacts as 'minor due to their effects.

From a review of the procedures followed for a number of offshore wind farm developments it was identified that a thorough knowledge of the prevailing physical conditions of the proposed site and relevant coastal regions is of fundamental importance in order to determine how the environment may be affected. The amount of information available to build this knowledge varies depending on the proposed site location. In certain cases information is already available regarding certain aspects of the physical environment e.g. tidal data, depth, wave conditions etc but in almost all situations field measurements or numerical modelling are required to fill important gaps in the data
Therefore a critical initial step in the examination of physical impacts is the typically desk based research that is required to identify and source all the relevant data that will enable an initial baseline evaluation of the site to be carried out. This data search is usually thorough and comprehensive given the high costs of collecting offshore environmental data and setting up and running appropriate numerical models. The nature of the initial evaluation is such that sufficient information regarding baseline conditions is attained in the first instance in order to determine the level of detail and priority areas in the impact assessment. As an example for certain Scottish Offshore wind farm developments detailed studies into sediment movement has been scoped out of overall impact assessment due to the nature of seabed (sediment type) and the generally benign wave and current conditions. Therefore proper and thorough baseline studies can help dictate the direction of the overall impact assessment and prevent unnecessary surveys and analysis being carried out.

Once the baseline condition has been established the procedure is usually relatively straightforward and involves modelling and empirical methodssuch that the various modelling and analysis show that the impacts of the proposed layout of the windfarm has on the baseline wave current and sediment transport processes. It is common conclusion from such studies that the wind turbine foundations given their typical size and spacings have negligible impact apart from scouring at the base for which mitigating measures can be provided. Ancillary works most notably cable laying generally has a low, local and short duration effect on sedimentary processes along the route of the cable. Given that cable laying is usually carried out during relatively calm conditions the potential for large scale seabed sediment mobilisation is not significant.

Wave energy Development

The wave energy industry has developed significantly in recent years and there are now a number of wave energy devices at a pre-commercial stage of development. Although the detail of the technologies is not the subject of this paper, a brief introduction four WECs, with different operational principles, is now provided in order to provide relevant background information.

Wavebob: This WEC is a self reacting point absorber designed for relatively deepwater deployments. There are two main parts which essentially are floats, one at the surface and another larger float located below the surface. When a wave acts on the device it induces a heave motion and causes the two floats to react against one another. This relative motion pressurises a hydraulic circuit which in turn drives an electrical generator. (Figure 1)

Wave Dragon: The Wave Dragon WEC is a potentially large device that uses wave run-up to elevate water above the sea surface. This water is then allowed to flow back through a number of low head hydro-turbines and thus generates electricity. The wave dragon has deflector arms that help focus the waves onto a ramp in order to maximise overtopping. A pre-commercial wave Dragon demonstrator has been planned to install of the UK coast at Milford Haven. This device has a rated capacity between 4-7MW and has a distance between arm tips of 300m. (Figure 2)

Pelamis: This WEC comprises of a number of cylindrical sections that are connected by hinged joints. When a wave acts on the device it causes relative motion between the adjacent sections which reacts against hydraulic cylinders and pump fluid into accumulators. This action results in the generation of electricity which is transferred to shore via seabed cables.

Oyster The Oyster WEC is being developed Aquamarine Power and is essentially a large flap that uses the surging motion of the waves to pump high pressure water to the shore where it drives a hydro-electric turbine. This device is usually located relatively close to the shoreline in shallow water (<15m)

Figure 1 Wavebob

Figure 2 Wave Dragon

Figure 3 Pelamis

Figure 4 Oyster

These devices will be referred to in later text as the magnitude of the impact is different depending on what type of device is deployed and its location relative to the shoreline.

Wave energy development has a number of commonalities with offshore wind but there are also essential differences which make the impacts potentially more significant.

- Wave energy convertors (WECs) take a portion of the incident energy out of the wave and can result in a reduction in height on the lee side of the device
- WECs can be much larger in scale as compared with wind turbine foundations e.g. Wave Dragon
- Tidal currents could be affected due to the interaction between the structure and the water
- Sediment transport regime will be affected if the wave and currents are changed from their pre-development magnitudes.

A thorough environmental assessment was carried out in relation to the Wave Hub test site in the UK. The steps followed in the methodology are summarised as follows (Halcrow (2006)),

1. To identify the environmental receptors/resources likely to be affected by the implementation of the proposed scheme through data collection, baseline surveys and consultation.
2. To identify the value or sensitivity of the environmental receptors/resources according to a five-point scale (i.e. very high, high, medium, low, and negligible), where appropriate.
3. To consider the interactions of the Wave Hub project with the existing environmental receptor/resource conditions to identify the potential impacts (i.e. changes) as a consequence of the Wave Hub project during construction, operation and decommissioning
4. To predict the magnitude of the potential impacts on environmental receptors/resources using predefined criteria. Magnitude was quantified on a four point scale (i.e. negligible, minor, moderate or major, including the scale (i.e. large to small) and nature (i.e. positive or negative) of the impact.
5. To quantify the significance of the potential effects on environmental receptors/resources based on the value/sensitivity and magnitude of the potential impacts and with due consideration of the duration (temporary or permanent) and reversibility of the impacts.
6. To identify mitigation and/or enhancement measures to avoid, reduce or remedy environmentally unacceptable or unnecessary adverse impacts to environmentally acceptable levels.
7. To identify the potential residual impacts of the Wave Hub project after mitigation measures are successfully implemented.

In the assessment of potential impacts at the Wavehub site a number of worst case scenarios for different placements of WECs were developed. For coastal processes it was the placement of 4 Wave Dragon devices that were considered to potentially give the most impact. After establishment of the baseline conditions and the use of various hydrodynamic and sediment transport models the following conclusions were obtained.

- Up to a 5% magnitude reduction in wave heights at the coast during a 1 in 1 year return period storm. For the typical case WEC layout, modelling predicts up to a 3% magnitude reduction in wave heights at the coast during a in 1 year return period storm.
- Wave modelling results show that wave heights are reduced in the sheltered sea area behind the WEC devices. As wave energies can propagate through the (semi-) floating WEC devices, waves can re-build after a distance of about 5km, depending upon the width and nature of the WEC devices.
- Under the worst case WEC layout scenario, the impact of the WEC device array layout with four west-facing Wave Dragon devices during mean wave conditions will be a reduction in wave heights at the coast of up to 13%.
- For the example small and big surfing wave conditions, modelling predicts a reduction in wave heights at the coast of up to 11%

- Under the typical case WEC layout scenario, the impact of the various WEC types during mean wave conditions will be a reduction in wave heights at the coast of up to 5%.
- The results of the sediment transport modelling for the WEC device layouts (worst case and typical case layout scenarios) are similar, but the worst case scenario shows changes over a larger area. This impact occurs because: The current flow patterns with or without the WECs are almost the same; and although the proposed WEC devices have an impact on wave patterns, this effect is not discernable at the seabed in water depths of more than 50m.
- For both WEC device layouts considered in the assessment, the predicted change to the wave climate and currents will not result in a discernable effect on sediment transport and beach levels along the north Cornish coast.
- No long term impacts would result from cable burial, provided the cable is buried to a sufficient depth to prevent future exposure.

The Environmental Impact Assessment (EIA) for the proposed Wave Dragon Pre-Commercial Demonstrator, located approximately 1.7km off the Pembrokeshire Coast at Long Point, examined physical impacts through all potential phases of the project. The study started with an examination of the baseline conditions and then proceeded to examine the response of the coastal system to the installation and operation of the WEC. Mitigation measures were proposed for any impacts that were considered significant. The assessment considered water and sediment quality, coastal processes (sediment fluxes, waves and tidal currents) and the onshore physical environment. The main findings are summarised below,

- The area was found to contain only small quantities of mobile sediment in the offshore environment and deployment area. Being a high energy site, it was concluded that most waterborne pollutants will be readily dispersed.
- Regarding coastal processes the study concluded that the proposed project would moderately impact on the waves close to the device and some localised impact on the currents and sedimentary processes in the immediate vicinity of the development. However, it concluded that there is unlikely to be any noticeable impacts further away from the device towards the shore.
- Regarding the onshore physical environment the assessment concluded that the geology and hydrology of the area could be sensitive to construction but by the application of best practice and environmental management controls during the construction phase, the project should have no adverse affects.

It should be noted that desk studies and numerical models were the primary methods used to examine impacts.

Physical Environment Assessment Methodology

Wave energy deployment locations are likely to be at locations with a mean annual power of at least 30kw/m and during standard operation WECs will take a portion of this energy resulting in lower wave heights on the lee side of the device. Typically WECs have efficiencies of less than 30% so most of the energy passes the device and continues to the shore. In addition as waves moves into shallower water, effects such as bed friction, refraction and breaking naturally dissipate the incident energy so in general if the WECs are located far enough offshore they should have a very negligible effect on the coastal environment. In addition most WECs are designed to operate for the mean wave conditions and go into survival mode during storm events. As sediment transport processes on the coastline are mostly driven by the larger storms then in such situations WECs should have a negligible impact. The main point here being that the impact studies undertaken must be designed specifically for the type and scale of the WEC deployment and the nature of the adjacent shoreline. The following is the methodology that is proposed for the assessment of the Irish test site at Belmullet, co. Mayo.

1. ***Initial Appraisal***; Study of site characteristics, nature of development, number and type of devices and operations required. This is essentially a desk study whereby an appraisal of the site is undertaken and potential sensitivities are identified. For instance the coastal area could be primarily rocky with very little fine sediments and in such a situation the scale of the study can be much reduced. Another example is if a number of Pelamis devices were being placed far offshore then the impact would be much less than that of a Wave Dragon device close to the shore. It is this type of assessment that makes up the opening phase of the study.
2. ***Data Gathering***; This is a more far reaching desk study whereby all relevant information pertinent to the site is collected. Some information will already be available as the developer is likely do have undertaken an examination of the site prior to selecting it. In addition there is likely to be large scale numerical model output in relation to waves and tidal currents readily available.
3. ***Scoping***; is defined by the European Commission (2001) as "the process of determining the context and extent of the matters which should be covered in the environmental information to be submitted to a competent authority for projects which are subject to EIA." It is the process of examining the gathered data and utilising it to identify potential impacts likely to be caused by the proposed development. An assessment is made regarding further investigative works in order to fill the gaps left in the existing data sets. The physical impacts should then be identified and classified similar to the manner as was undertaken in the Wave Hub study. Finally a scoping report is prepared which is essentially the plan for the EIA process.
4. ***Establish Baseline***; The initial part of this phase may involve fieldwork and the latter stages require numerical modelling. The fieldwork can be quite extensive requiring wave and tide measurement, sediment sampling, borehole logging, geophysical surveys etc. In certain cases the field work can take up to a year to complete due to difficulties in working at sites with high wave energy. Normally the field work will provide data for the calibration and validation of numerical models and will enable, along with the other data analysis work, a baseline condition for the site to be established. A baseline is not the system response to one particular set of environmental forcings but is an understanding, partly quantitative and partly qualitative, of how the coastal system will react to various types of inputs. As an example waves from a particular direction could generate longshore drift on an open beach and cross shore drift on pocket beaches in the designated zone of influence of the wave farm.
5. ***Assessment***; The assessment stage considers a number of scenarios in terms of wave farm layout and input environmental forcings and examines the sensitivity of the coastal system as compared to the baseline condition. This stage of the process involves numerical models that can show the extent to the change in wave and current conditions caused by the development. Variations in the sediment transport can also be determined but these only can be used on a qualitative a.
6. ***Mitigation***; This refers to the process of dealing with negative impacts that arise in the Assessment stage. Generally mitigation techniques available to the developer can involve one of the following; avoidance, reduction, compensation and remediation. Depending on the type of mitigation measures that are proposed the evaluation process will have to loop back to the assessment stage to determine if they would be effective. In some cases a few iterations will be required.
7. ***Conclusion***; This generally involves the preparation of the EIA report that outlines the work as described above and submitting it to the relevant authority for approval.

Belmullet Wave Energy Test Site

The Sutainable Energy Authority of Ireland (SEAI) is developing a National Wave energy test site off the coast of Annagh Head, near Belmullet, co. Mayo. It will be an open ocean, grid connected test

facility for pre-commercial technologies and will have an operational period of 15 years. As shown in figure 5 below the site has three different test areas each designed to cater for different types of WECs. Test site A which is approximately 15km offshore in approximately 100m water depth is for deepwater wave energy convertors (Wavebob, Pelamus etc.), Test site B located 5m from shore and in 50m water depth is designed for mid water/coastal wave energy convertors (Pelamus, OE Buoy etc.) whilst Test site C which is less than 1 km from shore and generally has a water depth less than 20m is for nearshore devices (Oyster) It is envisaged that all cables will be run ashore onto Beltarra Strand with an electrical sub-station being located at the rear of the strand.

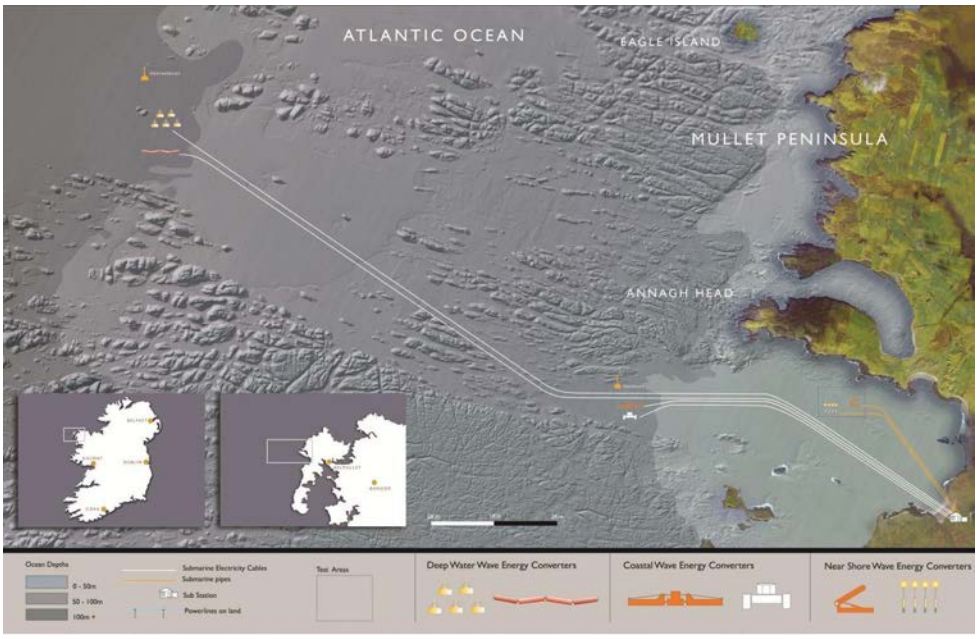

Figure 5 Belmullet wave energy Test Site (Ascoop 2010)

The study of the impacts on coastal processes was due to be completed in early 2010 but delays due to loss of equipment on the site means that the work will not now be completed until end September 2011. Therefore at time of writing it is not possible to provide any detail of the work only that it is following the steps that have been outlined in this document. It is expected that more detailed information will be provided in the conference presentation.

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Contact author: Dr. Jimmy Murphy

Hydraulics and Maritime Research Centre
Pouladuff Road
University College Cork
Ireland
Tel: +353 21 4250023
Email: jimmy.murphy@ucc.ie.